

Spatial Domain Identification Methods and Domain Shape Analysis

MGC Spatial course
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12th November 2025

Spatial Domain Identification Methods and Domain Shape Analysis

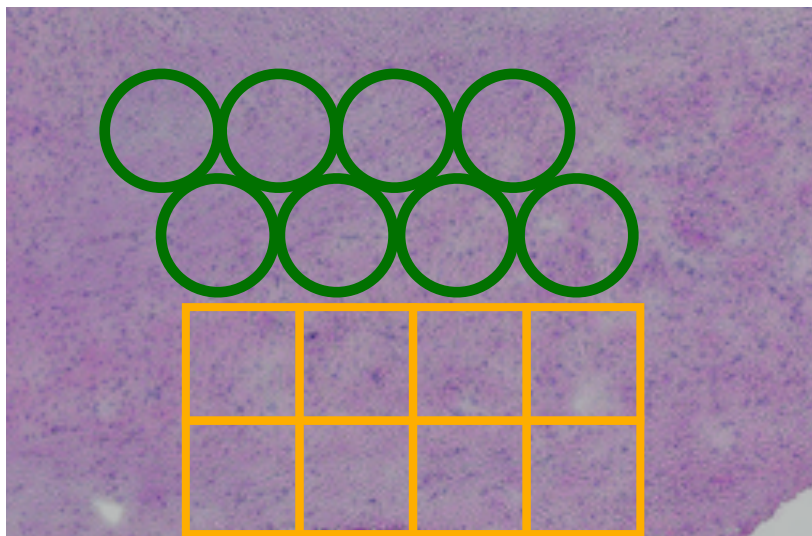
Outline:

- 1.(Distance-based) graph construction
- 2.Spatial clustering methods
 - a. Banksy
 - b. CellCharter
- 3.Domain analysis

Graph construction

Differs in sequencing-based and imaging-based spatial datasets:

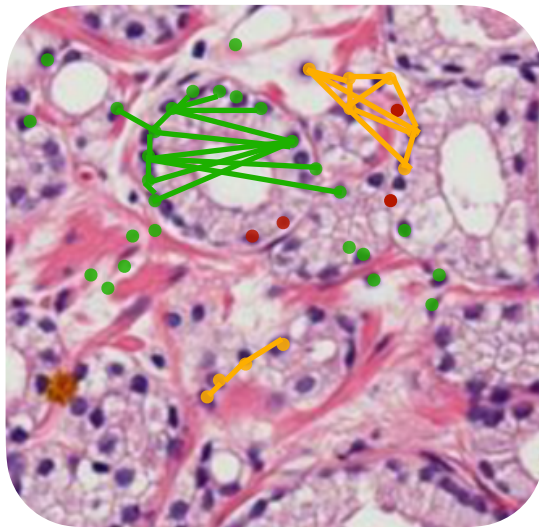
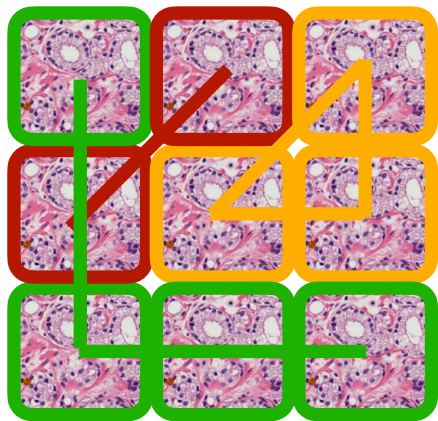
- Sequencing-based = spatial units are organised in a grid- or hexagon-structure



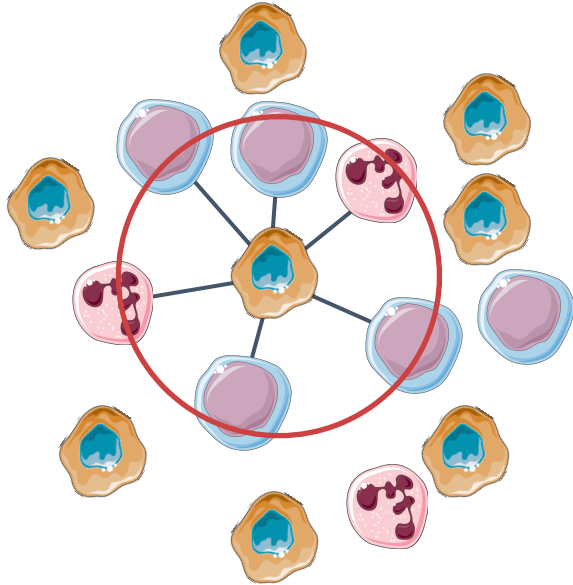
Graph construction

Differs in sequencing-based and imaging-based spatial datasets:

- Sequencing-based = spatial units are organised in a grid- or hexagon-structure
- Imaging-based = single cell resolution = cells distances can be used



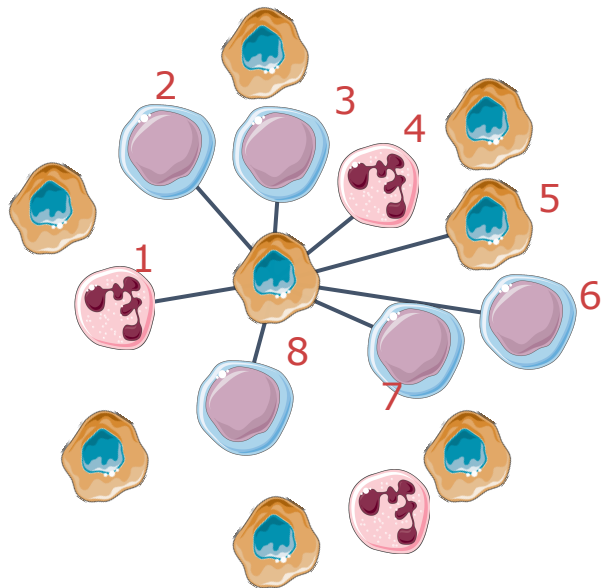
Graph construction



Find connectivities:

- Radius-based = all the spatial units within a distance
- Closest neighbors (e.g. k-nearest neighbors) = set number of nearest neighbours
- Step-wise (n steps)

Graph construction



Find connectivities:

- Radius-based = all the spatial units within a distance
- Closest neighbors (e.g. k-nearest neighbors) = set number of nearest neighbours - **Banksy**
- Step-wise (n steps) - **CellCharter**

k=8

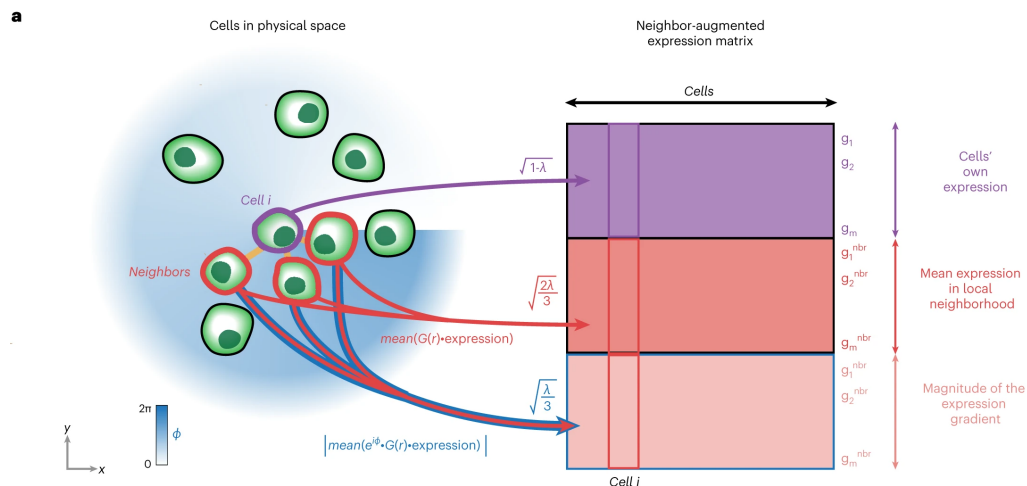
Banksy

Assumption: a cell's state can be more fully represented by considering both its own transcriptome and that of its local microenvironment.

Feature augmentation strategy

Concatenation of:

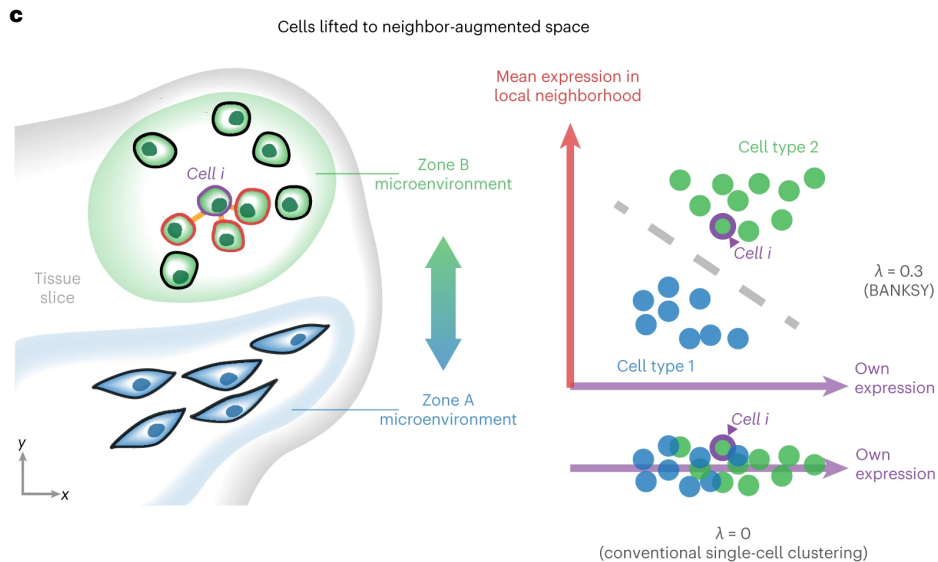
- Own gene expression
- Mean gene expression of local neighbourhood
- Magnitude of the expression gradient



Original size: n cells $\times$ g genes

Augmented size: n cells $\times$ $3 \times g$ genes

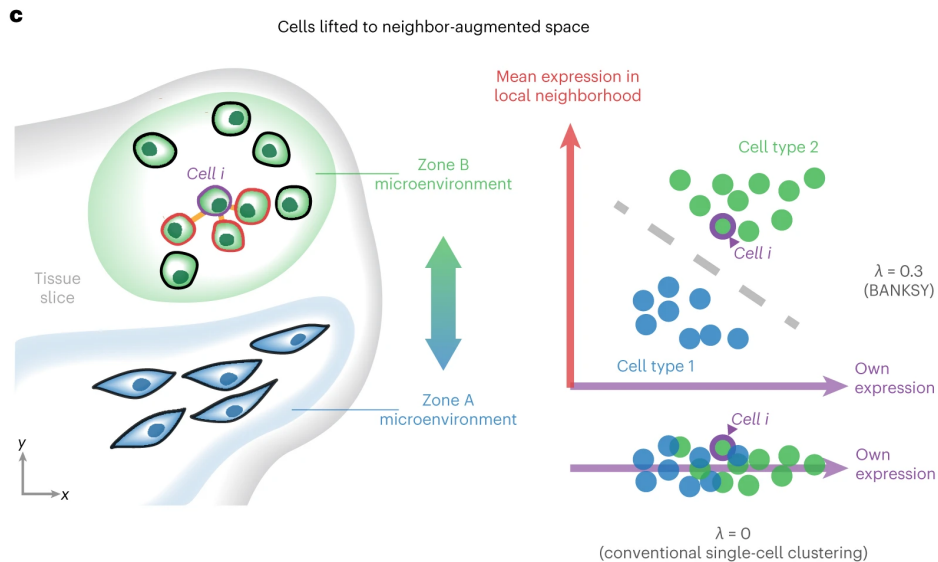
Banksy



● and ● have similar gene expression

-> no differences with conventional clustering

Banksy

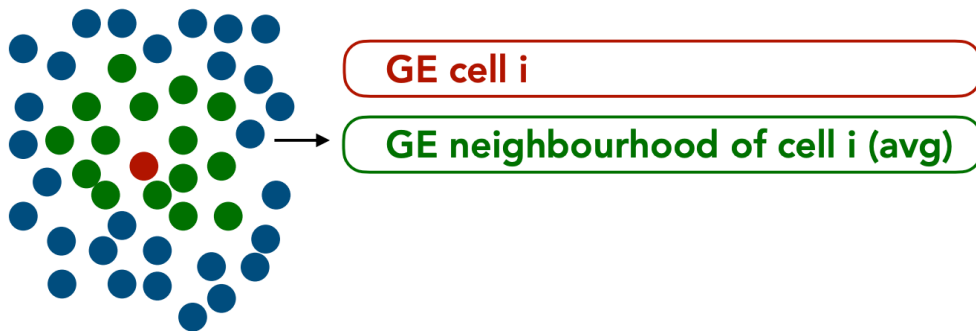


● and ● have similar gene expression

-> but differences are detected when the neighbourhood is taken into consideration

-> no differences with conventional clustering

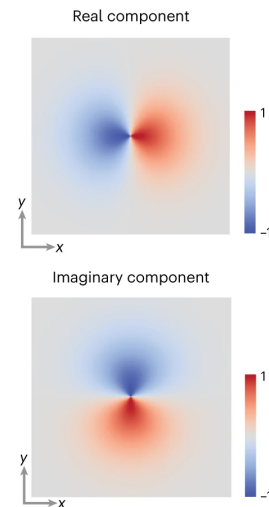
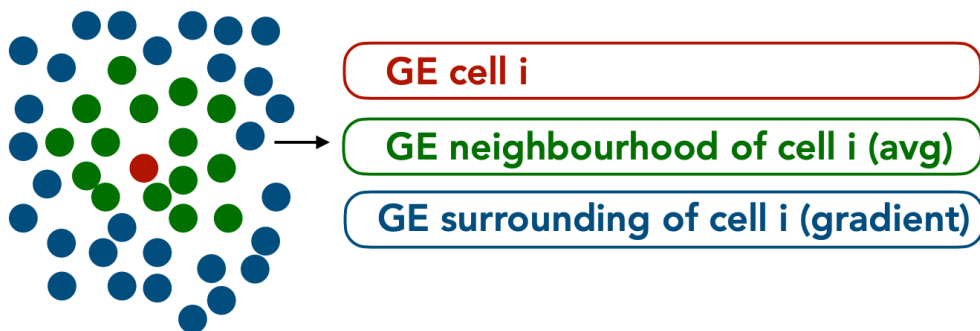
Banksy



From the original count matrix (matrix C):

- Construct a **neighborhood graph** between cells in physical space using k-nearest neighbors (Matrix M) and compute the mean expression in the neighborhood of a cell

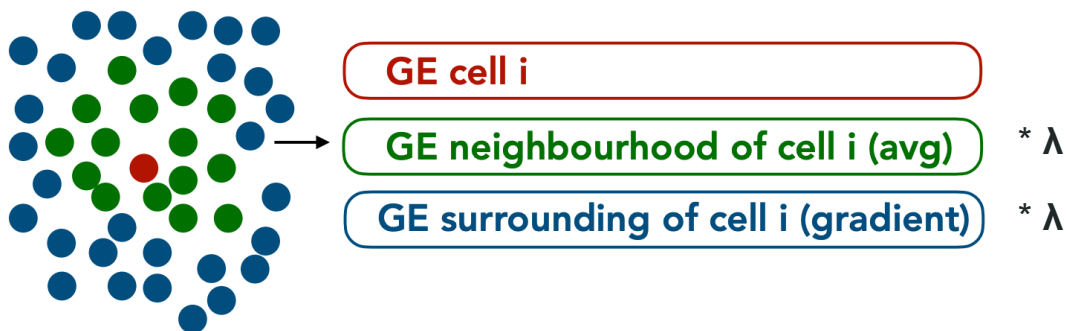
Banksy



From the original count matrix (matrix C):

- Construct a **neighborhood graph** between cells in physical space using k -nearest neighbors (Matrix M) and compute the mean expression in the neighborhood of a cell
- AGF (Azimuthal Gabor Filter)**: measures gradients in gene expression in each neighborhood (size= $2*k$). (Matrix G)

Banksy



From the original count matrix (matrix **C**):

- Construct a **neighborhood graph** between cells in physical space using k-nearest neighbors (Matrix **M**) and compute the mean expression in the neighborhood of a cell
- AGF**: measures gradients in gene expression in each neighborhood (size= $2 \times k$). (Matrix **G**)
- Matrices **M** and **G** are **scaled** based on a mixing parameter λ , which controls their relative weighting.
- Concatenate** matrices **M**, **G** and **C**.

Banksy

Parameters:

- Weighting parameter $\lambda \in [0, 1]$

Weights the **contributions** of the cell-transcriptome matrix and the neighbor expression matrices. **Smaller λ settings emphasize cells' own transcriptomes** (and cause cells to cluster according to cell type). In the limit, when $\lambda = 0$, BANKSY reduces to conventional nonspatial clustering.
- K-geom

Number of nearest neighbors to be included in the neighbourhood
 $2 * k\text{-geom} = \text{number of spatial units that will be calculated with the gradient}$
- Number of Principal Components
- Clustering resolution

Banksy

Step by step

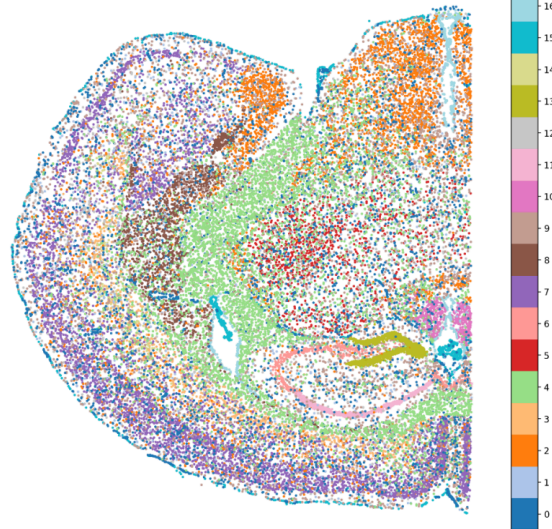
1. Normalise data
2. Construction of k-nearest-neighbors based on physical distance
3. Generate spatial weights from distance
4. Generate Banksy matrix = From the original count matrix (matrix C):
 - a. Construct a **neighborhood graph** between cells in physical space using k-nearest neighbors (Matrix M) and compute the mean expression in the neighborhood of a cell
 - b. **AGF**: measures gradients in gene expression in each neighborhood. (Matrix G)
 - c. Matrices M and G are scaled based on a mixing parameter λ , which controls their relative weighting.
 - d. **Concatenate** matrices M , G and C .
6. Dimensionality reduction (PCA)
7. Run clustering (Leiden)

Banksy

7. Visualise the results

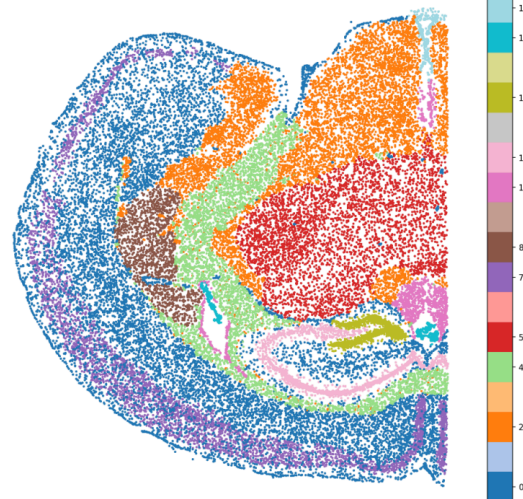
Lambda = 0

BANKSY Labels (nonspatial_pc20_nc0.00_r0.10)



Lambda = 0.6

BANKSY Labels (scaled_gaussian_pc20_nc0.60_r0.10)



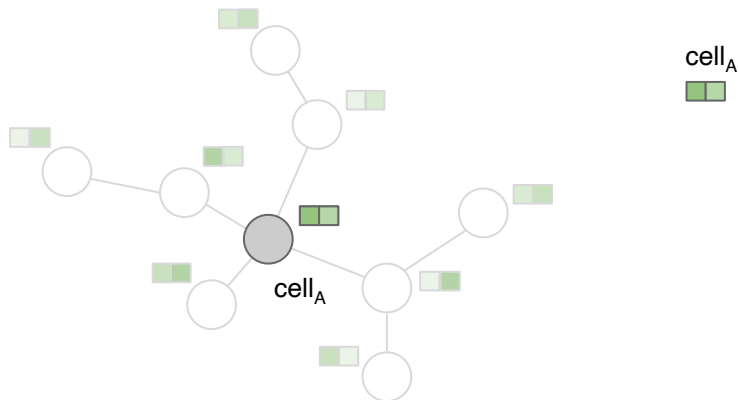
What will the spatial domains look like if lambda = 1?

What if k-geom = 100?

CellCharter

Step by step

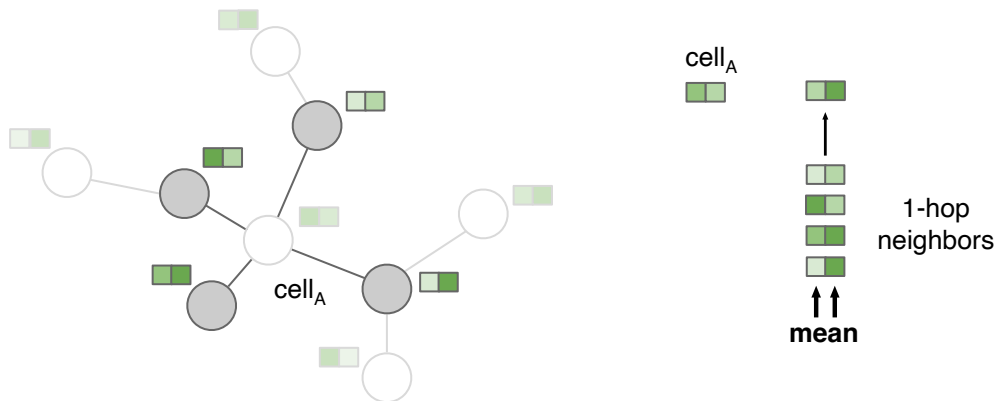
1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = 1 steps away * **with some checks for irregular structures**
 - a. Gene expression of cell A



CellCharter

Step by step

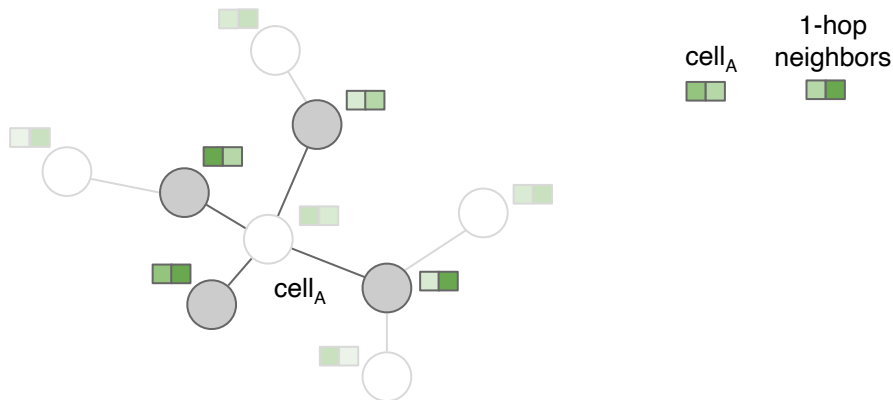
1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = 1 steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighborhood



CellCharter

Step by step

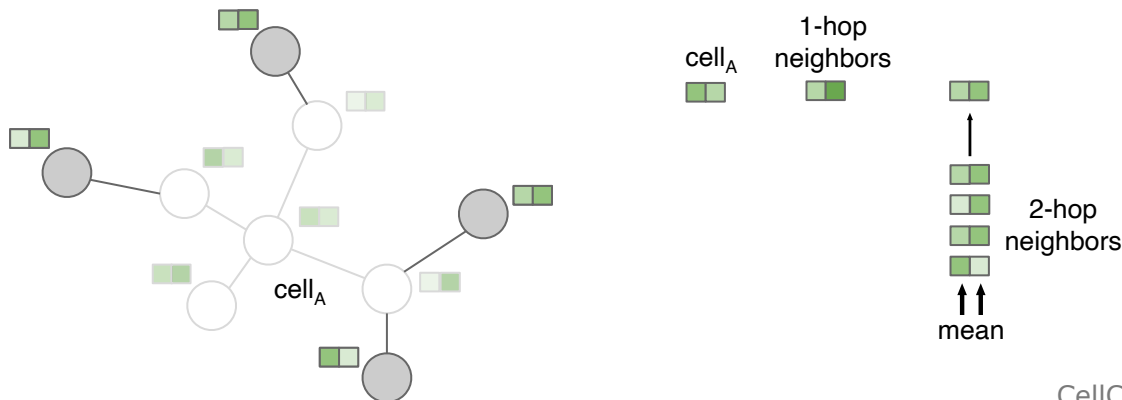
1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = 1 steps away
 - a. Gene expression of cell A
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CellCharter

Step by step

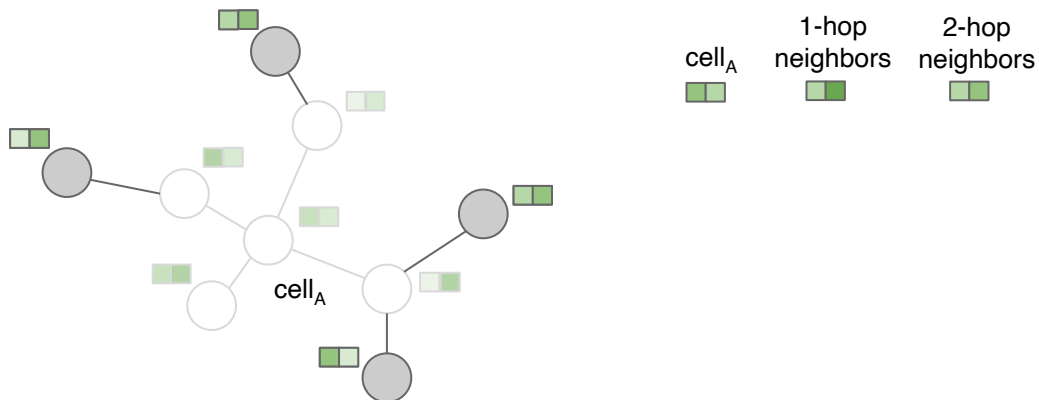
1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = 1 steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighbourhood
 - c. Mean gene expression of second neighbourhood



CellCharter

Step by step

1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = l steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighbourhood
 - c. Mean gene expression of second neighbourhood
 - d.
 - e. Mean gene expression of l -th neighbourhood

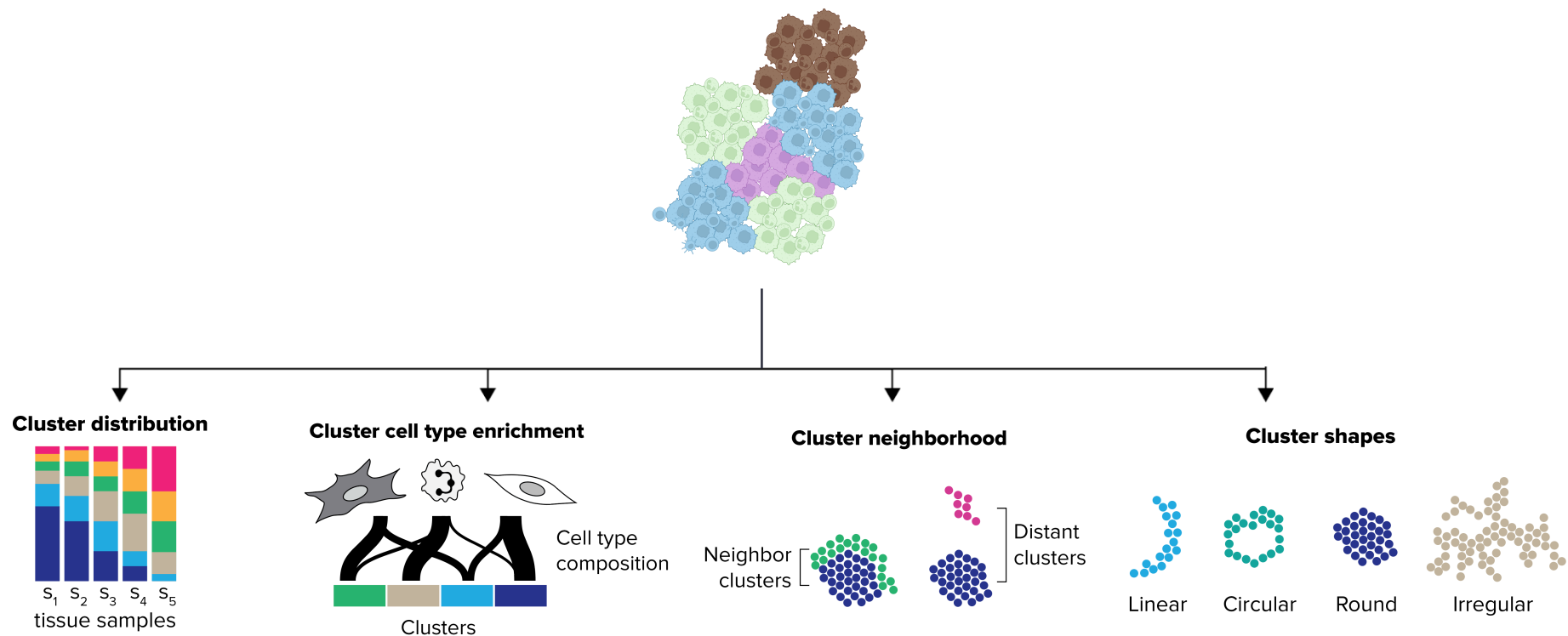


CellCharter

1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = l steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighbourhood
 - c. Mean gene expression of second neighbourhood
 - d.
 - e. Mean gene expression of l -th neighbourhood
3. Concatenation (neighbourhood aggregation method to obtain one feature vector from the concatenation of multiple feature vectors)
4. Clustering using a Gaussian mixture model (GMM)



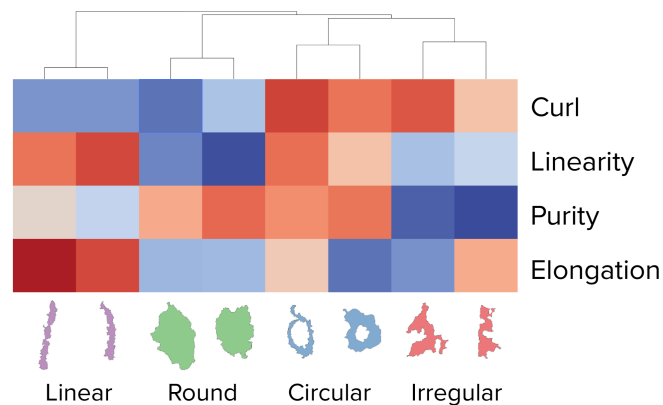
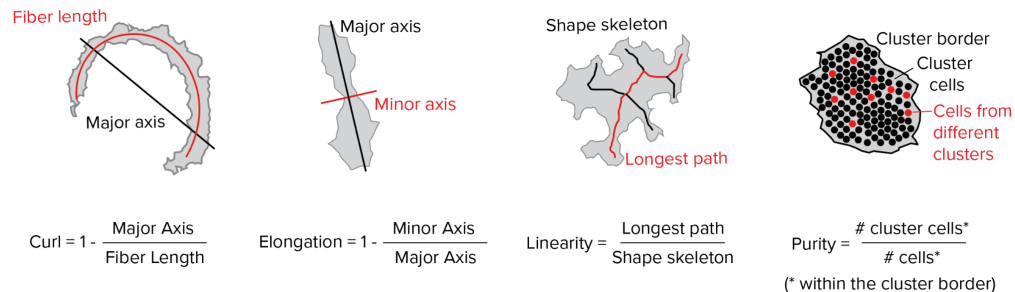
CellCharter



Transcriptomics-agnostic Spatial Domain (preliminary) analysis

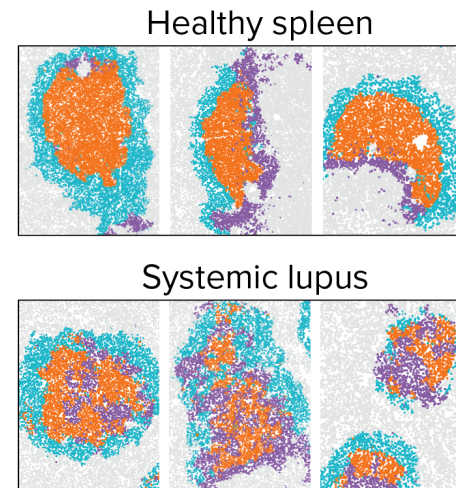
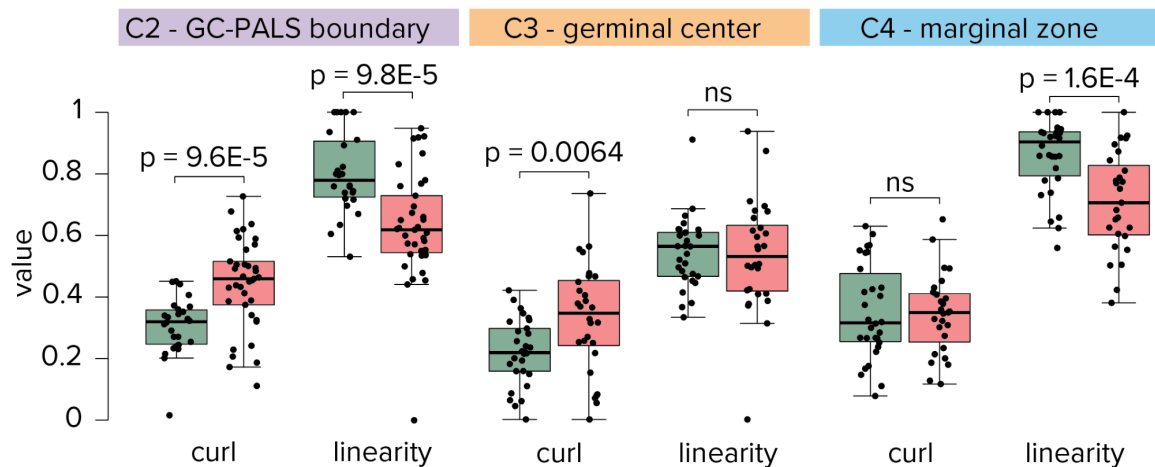
Shape metrics

- Curl: how curved or twisted a shape is
- Elongation: ratio of major and minor axis
- Linearity: how well a shape can be approximated by a liner path
- Purity: quantifies the fraction of cells within a cluster boundary that belong to that cluster



Transcriptomics-agnostic Spatial Domain (preliminary) analysis

Shape metrics

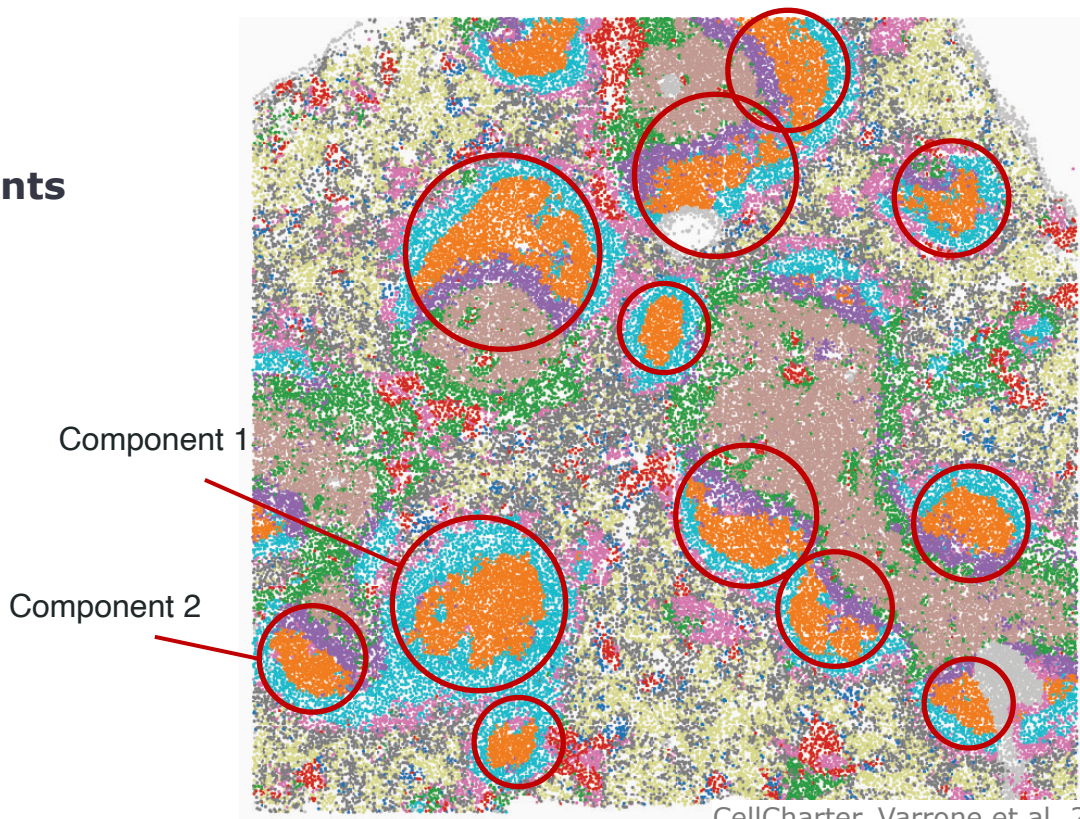


Transcriptomics-agnostic Spatial Domain (preliminary) analysis

Shape metrics

For each spatial domain

1. Split it into different **components**

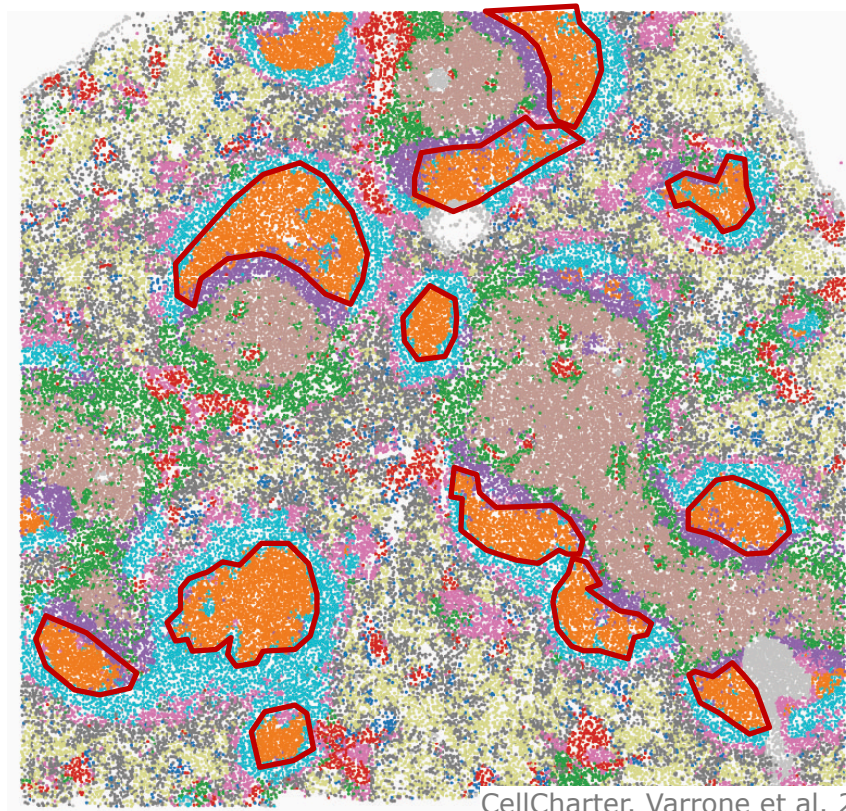


Transcriptomics-agnostic Spatial Domain (preliminary) analysis

Shape metrics

For each spatial domain

1. Split it into different **components**
2. Automatically draw the **boundary** around each components
 - *Using alpha shapes*

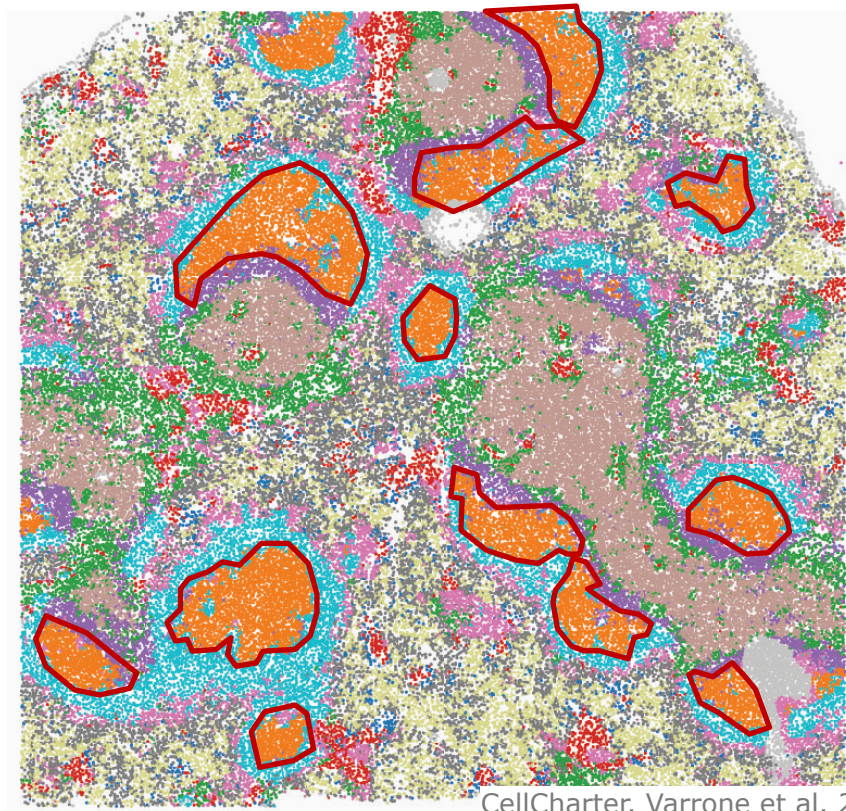


Transcriptomics-agnostic Spatial Domain (preliminary) analysis

Shape metrics

For each spatial domain

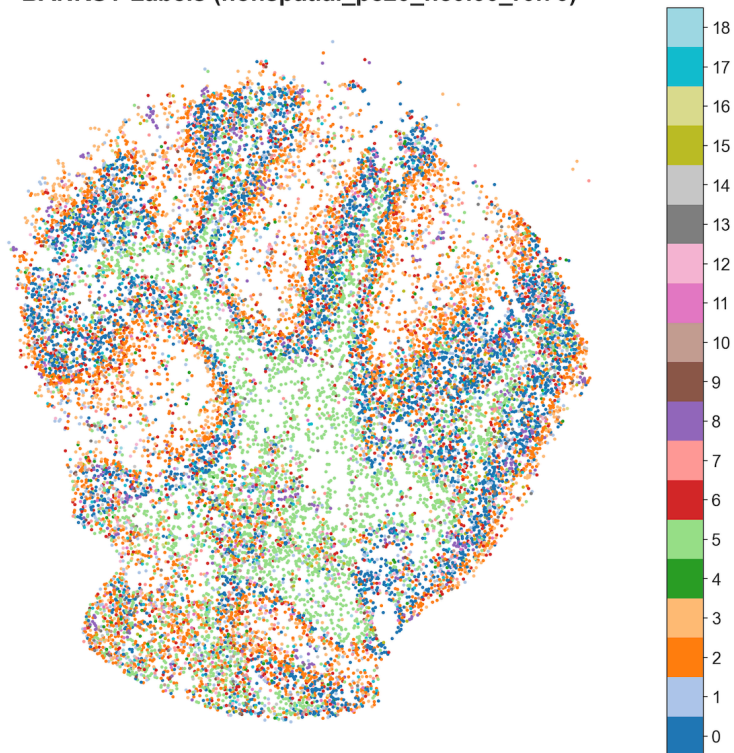
1. Split it into different **components**
2. Automatically draw the **boundary** around each components
- *Using alpha shapes*
3. Compute **shapes metrics** for each component



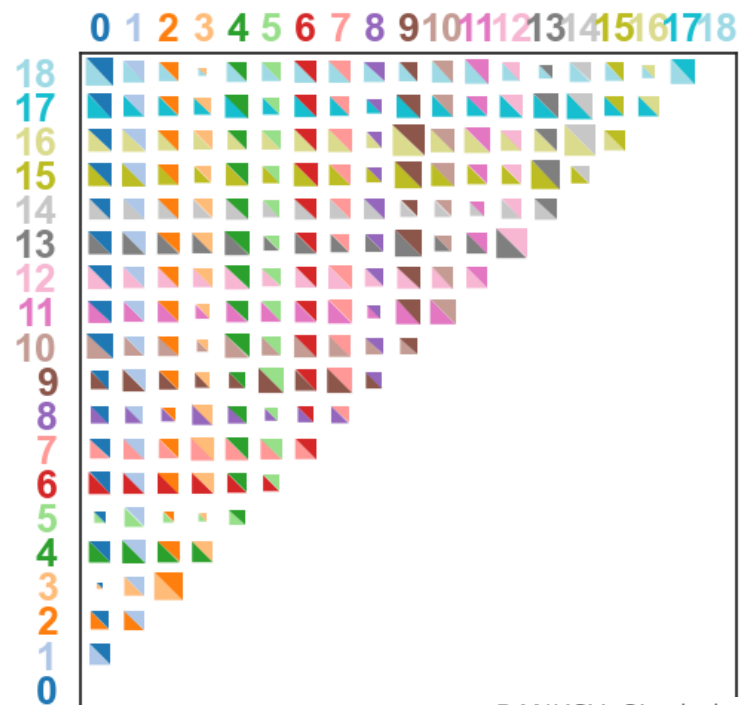
Transcriptomics-agnostic Spatial Domain (preliminary) analysis

Co-localization

BANKSY Labels (nonspatial_pc20_nc0.00_r0.70)



Connections between label



Transcriptomics-agnostic Spatial Domain (preliminary) analysis

Cell-type composition

B

